Defining General, Hypothetical Interventions

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Have you ever begun reading a paper in the methodological causal inference literature and encountered the phrase "assume the treatment or exposure is a binary..."? (Most papers we read assume this!!) While assuming exposure variables are binary can simplify the definition of causal effects, many exposures of interest in reality are not binary.

Instead, we will work in situations where A is a binary, categorical, multivariate, or continuous variable!

In the previous section, we defined $\mathsf{d}(a_t, h_t, \epsilon_t)$ as a function that maps A_t, H_t, and potentially a randomizer ϵ_t to a new value of A_t. Our focus henceforth is estimating the causal effect of an intervention, characterized by d on the outcome Y, through the causal parameter

$$\theta = \mathsf{E}[Y^{\bar{A}^{\mathsf{d}}}],$$

where $Y^{\bar{A}^{\mathsf{d}}}$ is the counterfactual outcome in a world, where possibly contrary to fact, each entry of $\bar{A} = (A_1, \ldots, A_\tau)$ was modified according to the function d.

- When Y is continuous, θ is the mean population value of Y under intervention d.
- When Y is dichotomous, θ is the population proportion of event Y under intervention d.
- When Y is the indicator of an event by end of the study, θ is defined as the cumulative incidence of Y under intervention d.

But what is this function d, how can it be defined, and how does using this function to define interventions solve the problem? Let's start from simple to more complex examples of functions d.

Static Interventions

Let A denote a binary vector, such as receiving a medication, and define $\mathsf{d}(a_t, h_t, \epsilon) = 1$. This intervention characterizes a hypothetical world where all members of the population receive treatment.

> An intervention is static if the function d_t always returns the same value regardless of the input.

Example

> Let's say we were interested in the effect of randomizing patients with opioid use disorder to injection naltrexone ($A = 1$) vs. sublingual buprenorphine ($A = 0$). We would contrast the counterfactual outcomes in a hypothetical world in which all units were treated with injection naltrexone $\mathsf{d}_1 = 1$ versus a hypothetical world in which all units were treated with buprenorphine $\mathsf{d}_0 = 0$. This gives us the well-known average (comparative) treatment effect (ATE).
>
> $$\mathsf{E}[Y^{\bar{A}^{\mathsf{d}_1}} - Y^{\bar{A}^{\mathsf{d}_0}}] = \mathsf{E}[Y^{A=1} - Y^{A=0}]$$

Dynamic Treatment Regime

Let A_t denote a binary vector, such as receiving a medication, and H_t a numeric vector, such as a measure of discomfort. For a given value of δ, define

$$\mathsf{d}(a_t, h_t, \epsilon) = \begin{cases} 1 & \text{if } h_t > \delta \\ 0 & \text{otherwise.} \end{cases}$$

> Interventions where the output of the function d depends **only on the covariates** H_t are referred to as being dynamic.

Example

> Rudolph, Williams, et al. (2022) examined a Buprenorphine (BUP-NX) dosing strategy among a population of patients who were taking BUP-NX for opioid use disorder. Under the hypothetical intervention, patients who reported opioid use during the week prior to a physicians exam received a BUP-NX dose increase while patients who did not report prior-week opioid use maintained the same dose. Let A_t be a binary indicator for BUP-NX dose increase at week t compared to week $t-1$ and X_t be an indicator for opioid use at week t. Then,
>
> $$\mathsf{d}(a_t, h_t, \epsilon) = \begin{cases} 1 \text{ if } x_{t-1} = 1 \\ 0 \text{ otherwise} \end{cases}$$

Modified Treatment Policies

While much attention is given to static and dynamic interventions, their use is often accompanied by a few key problems.

1. **Defining causal effects in terms of hypothetical interventions where treatment is applied to all units may be inconceivable.** For example, we may be interested to know if reducing surgery time reduces surgical complications. However, it's inconceivable to set all surgeries to a given duration, even if this duration depends on patient covariates.

2. **Defining causal effects in terms of hypothetical interventions where treatment is applied to all units may induce positivity violations.**

A solution to these problems is to instead define causal effects using **modified treatment policies (MTP).**

> An intervention characterized by a hypothetical world where the *natural* value of treatment is modified is called a *modified treatment policy*.

Additive and multiplicative shift MTP

Let A_t denote a numeric vector. Assume that A_t has support in the data such that $P(A_t \leq u(h_t) \mid H_t = h_t) = 1$. For an analyst-defined value of δ, define

$$\mathsf{d}(a_t, h_t, \epsilon) = \begin{cases} a_t + \delta & \text{if } a_t + \delta \leq u(h_t) \\ a_t & \text{otherwise.} \end{cases}$$

Under this intervention, the natural value of exposure at time t is increased by the analyst-defined value δ, whenever such an increase is feasible. This MTP is referred to as an *additive shift MTP*.

Example

> Díaz et al. (2023) estimated the effect of increasing P/F ratio (a measure of hypoxemia) by 50 units on survival among those patients with acute respiratory failure (a P/F ratio < 300).
>
> $$\mathsf{d}_t(a_t, h_t, \epsilon) = \begin{cases} a_t + 50 & \text{if } a_t \leq 300 \\ a_t & \text{otherwise} \end{cases}$$

We can similarly define a *multiplicative shift MTP* as

$$\mathsf{d}(a_t, h_t, \epsilon) = \begin{cases} a_t \times \delta & \text{if } a_t \times \delta \leq u(h_t) \\ a_t & \text{otherwise.} \end{cases}$$

Example

> Nugent and Balzer (2023) evaluated the association between county-level measures of mobility and incident COVID-19 cases in the United States in the Summer and Fall of 2022. They considered both

> hypothetical additive and multiplicative MTPs; for example, they defined a multiplicative MTP where a measure for the density of mobile devices visiting commercial locations was decreased by 25%:
>
> $$\mathsf{d}(a_t, h_t, \epsilon) = a_t \times 0.75.$$

Randomized Interventions

Let A denote a binary vector, $\epsilon \sim U(0,1)$, and ϵ be an analyst-defined value between 0 and 1. We may then define randomized interventions. For example, imagine we are interested in a hypothetical world where half of all smokers quit smoking. This intervention would be defined as

$$\mathsf{d}(a_t, \epsilon_t) = \begin{cases} 0 & \text{if } \epsilon_t < 0.5 \text{ and } a_t = 1 \\ a_t & \text{otherwise} \end{cases}.$$

Incremental Propensity Score Interventions Based on the Risk Ratio

Let A denote a binary variable, $\epsilon \sim U(0,1)$, and δ be an analyst-defined risk ratio limited to be between 0 and 1. In addition, define $P(A_t = a_t \mid H_t) = \mathsf{g}(a_t \mid H_t)$.

If we were interested in an intervention that decreased the likelihood of receiving treatment, define

$$\mathsf{d}_t(a_t, h_t, \epsilon_t) = \begin{cases} a_t & \text{if } \epsilon_t < \delta \\ 0 & \text{otherwise} \end{cases}.$$

In this case, we have $\mathsf{g}^{\mathsf{d}}(a_t \mid H_t) = a_t \delta \mathsf{g}_t(1 \mid H_t) + (1 - a_t)(1 - \delta \mathsf{g}_t(1 \mid H_t))$, which leads to a risk ratio of $\mathsf{g}_t^{\mathsf{d}}(1 \mid H_t)/\mathsf{g}_t(1 \mid H_t) = \delta$ for comparing the propensity score post- vs pre-intervention.

> An intervention where the conditional probability of treatment is shifted is referred to as an incremental propensity score intervention.

Conversely, if we were interested in an intervention that increased the likelihood of receiving treatment, define

$$\mathsf{d}_t(a_t, h_t, \epsilon_t) = \begin{cases} a_t & \text{if } \epsilon_t < \delta \\ 1 & \text{otherwise.} \end{cases}$$

Now $\mathsf{g}_t^{\mathsf{d}}(a_t \mid H_t) = a_t(1 - \delta \mathsf{g}_t(0 \mid H_t)) + (1 - a_t)\delta \mathsf{g}_t(0 \mid H_t)$, which implies a risk ratio $\mathsf{g}_t^{\mathsf{d}}(0 \mid H_t)/\mathsf{g}(0 \mid H_t) = \delta$.

> Interventions where the shift is in the odds ratio scale were previously proposed, but the effects of odds-ratio shifts should not be estimated with `lmtp`, we will discuss this more later.

Example

> Using electronic health record data, Wen, Marcus, and Young (2023) estimated the effect of increasing the proportion of PrEP uptake on bacterial STI among cis-gender males being tested for STIs and that do not have HIV. Let A_t be a binary indicator for PrEP initiation at week t, and L_t be a binary indicator for any STI testing and being HIV-free at week t. They defined a "medium" successful PrEP uptake intervention as
>
> $$\mathsf{d}(a_t, h_t, \epsilon) = \begin{cases} a_t & \text{if } l_t = 1 \text{ and } \epsilon_t < 0.85 \\ 1 & \text{otherwise.} \end{cases}$$

Identification of the causal parameter

Recall that the fundamental problem of causal inference is that we can't observe the alternative worlds which we use to define causal effects. If we can't observe counterfactual variables, then how can we learn a causal effect? Under a set of certain assumptions, we can *identify* a causal parameter from observed data. These assumptions are called identification assumptions.

Identification Assumptions

1. *Positivity*. If $(a_t, h_t) \in \text{supp}\{A_t, H_t\}$ then $\mathsf{d}(a_t, h_t) \in \text{supp}\{A_t, H_t\}$ for $t \in \{1, \ldots, \tau\}$.

   > If there is a unit with observed treatment value a_t and covariates h_t, there must also be a unit with treatment value $\mathsf{d}(a_t, h_t)$ and covariates h_t.

2. *No unmeasured confounders*. All the common causes of A_t and (L_s, A_s, Y) are measured and contained in H_t for all $s \in \{t+1, \ldots, \tau\}$.

   > For all times t, the history H_t contains sufficient variables to adjust for confounding of A_t and any subsequent variables, including future treatment.

> **Question**: When might these assumptions be violated?

Assuming the above, θ is identified from the observed data with:

Set $\mathsf{m}_{\tau+1} = Y$ For $t = \tau, \ldots, 1$, recursively define

$$\mathsf{m}_t : (a_t, h_t) \to \mathsf{E}[\mathsf{m}_{t+1}(A_{t+1}^{\mathsf{d}}, H_{t+1}) \mid A_t = a_t, H_t = h_t],$$

and define $\theta = E[\mathsf{m}_1(A_1^{\mathsf{d}}, L_1)]$.

Example

Consider the following data where $\tau = 2$.

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L1	A1	L2	A2	Y
-0.10	1	0.30	1	1.54
0.40	0	-0.74	0	0.05
0.36	1	-1.27	0	2.71
1.10	1	0.51	1	0.25
-2.27	0	-0.47	0	3.04
0.00	1	-0.64	1	2.53
-1.30	0	0.91	1	2.16
0.43	0	1.11	1	-3.14
-2.15	0	2.11	1	-2.37
-0.30	0	-2.11	0	4.44

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We can compute the identification formula in the following steps:

1. Set $\mathsf{m}_3(A_3^{\mathsf{d}}, H_3) = Y$

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```
m3_d <- foo$Y
```

2. Compute the regression of $\mathsf{m}_3(A_3^{\mathsf{d}}, H_3)$ on (A_2, H_2). This gives a predictive function, call that predictive function $\mathsf{m}_2(A_2, H_2)$.

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```
m2 <- glm(m3_d ~ L1 + A1 + L2 + A2,
            data = dplyr::rename(foo, m3_d = Y))
```

3. Use the predictive function to compute what would have occurred if the intervention had been implemented at time $t = 2$, i.e., compute $\mathsf{m}_2(A_2^{\mathsf{d}}, H_2)$.

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```
m2_d <- predict(m2, dplyr::mutate(foo, A2 = 1))
```

4. Compute the regression of $\mathsf{m}_2(A_2^{\mathsf{d}}, H_2)$ on (A_1, H_1). This gives a predictive function, call that predictive function $\mathsf{m}_1(A_1, H_1)$.

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```
m1 <- glm(m2_d ~ L1 + A1, data = dplyr::mutate(foo, m2_d = m2_d))
```

5. Use the predictive function to compute what would have occurred if the intervention had been implemented at time $t = 1$, i.e., compute $\mathsf{m}_1(A_1^{\mathsf{d}}, H_1)$.

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```
m1_d <- predict(m1, dplyr::mutate(foo, A1 = 1))
```

6. Compute the mean of $\mathsf{m}_1(A_1^{\mathsf{d}}, H_1)$. This mean is equal to θ.

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```
mean(m1_d)
```

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